

MUSTAPHA ABDULLAHI

Espoo, Finland

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RESEARCH INTERESTS

Machine Learning Systems, Reinforcement Learning, and Large Language Models

EDUCATION

Aalto University & Finnish Center of Artificial Intelligence (FCAI) **April 2024 - Present**
PhD Artificial Intelligence *Espoo, Finland*

Advisor: **Prof. Bo Zhao**

Queen Mary University of London **Sept. 2022 - Sept. 2023**
MSc Artificial Intelligence [Computer Vision and Robotics] - Grade: 80% (Distinction) *London, U.K.*

Supervisor: **Prof. Bo Zhao**

Ahmadu Bello University **Oct. 2015 - Nov. 2021**
BEng Computer Engineering - CGPA: 4.90/5.00 (First-class) *Zaria, Nigeria*

Supervisor: **Prof. Habeeb Bello-Salau**

Federal Government College, Kaduna **Sept. 2012 - July 2015**
West African Secondary School Certificate (W.A.S.S.C.E) *Kaduna, Nigeria*

MINOR EDUCATION

Mediterranean Machine Learning Summer School (M2L School), Thessaloniki, Greece. **Aug. - Sept. 2023**

The Cornell, Maryland, Max Planck Pre-doctoral Research School (CMMRS), Germany **Aug. 2023**

Reinforcement Learning Summer School (RLSS), Barcelona, Spain **June - July 2023**

INDUSTRY EXPERIENCE

QuantumBlack AI Labs (McKinsey & Company) **Oct. 2023 - Jan. 2024**
Software Engineer Intern *London, U.K.*

- Developed a one-click CLI tool for effortless deployment of **Kedro** projects on **Databricks** Platform.
- Implemented essential client study feedback as integral components within software production pipelines.

Play4Work **Feb. 2021 - March 2022**
Chatbot Developer *Canada(Remote)*

- Collaborated with a team of 5 to design conversation mappings required to build an interactive chatbot.
- Spearheaded a team of 2 to develop a chatbot with over 700 conversation nodes with Google DialogFlow.
- Reduced 60% of the company's workload of candidate's matching and responses processing.

Huawei **Jan. 2020 - Feb. 2020**
Big Data/Cloud Intern *Lagos, Nigeria*

- Set up and managed FTP Server, WordPress (Windows & Linux), Windows Share Point Server, etc. on Huawei Elastic Cloud Server (ECS).
- Performed Relational Database migration using Cloud Data Migration Tool. Utilised Squoop to migrate data between Apache HDFS, Apache HBase, Apache Hive and MySQL database.
- Performed Data Analysis of Smart IoV solution with over 400,000 data points utilizing HiveQL for offline analysis, Apache HBase for real-time query and retrieval and Spark-SQL to migrate the offline analysis service.

RESEARCH EXPERIENCE

Large-Scale Data & Systems (LSDS), Imperial College London **June 2023 - Aug. 2023**
Research Intern [Master's Thesis] (Advised by: Prof. Bo Zhao & Prof. Peter Pietzuch) *London, U.K.*

- Evaluated tradeoffs among three distribution policies of MindSpore Distributed RL.

AWARDS & ACCOLADES

- Finnish Doctoral Program Network in Artificial Intelligence (AI-DOC)
funded by Finnish Center of Artificial Intelligence (FCAI). 2024-2027
- Finland Fellowship [for doctoral studies at Aalto University] 2024
- Black in AI's Emerging Leaders in AI (ELAI) 2023
- AEGEAN Scholarship - Travel Grant to attend M2L School (one of 10 top students) 2023
- Best Graduating Engineering Student Award, NSE (Nigerian Society of Engineers) 2022
- 2023 Rhodes Scholarship Finalist [West Africa] 2022
- Google DeepMind Scholarship 2022-2023
- QuantumBlack (McKinsey Company)/Office of AI Scholarship 2022-2023
- i-Scholar Initiative (ISI) Awardee 2022
- Best ICT Competition Instructor, Huawei ICT Competition Sub-Saharan Africa 2022
- 1st Place, Dept. of Computer Eng. Final Year Undergraduate Project Exhibition 2021
- 3rd Best Engineering Graduate in Nigeria (Awarded by Committee of Deans of Engineering and Technology of Nigerian Universities (CODET)) 2021
- NCS (Nigerian Computer Society) Jim Ovia Scholarship Award 2020-2021
- Grand Prize Winner, Huawei ICT Competition Global Final - Cloud Track 2020
- 1st Place, Huawei ICT Competition Regional Final (Southern Africa) - Cloud Track 2020
- 3rd Place, Huawei ICT Competition National Final - Cloud Track 2019
- Nigeria Higher Education Foundation Scholar (Class of 2019) 2019
- 2nd Runner Up, Professor Ayodele Awojobi Design Competition (PAADC) 3.0 2019
- NNPC/Total National Merit Scholarship Award 2018-2021
- China-Nigeria October 1st Scholarship 2018
- Federal Government Scholarship Award [Undergraduate Award] 2017-2021
- Agbami Special Medical and Engineering Professionals Scholarship Award 2017-2021

PUBLICATIONS

- [1] Zheyue Tan, **Mustapha Abdullahi**, Tuo Shi, Huining Yuan, Zelai Xu, Chao Yu, Boxun Li, and Bo Zhao, "EARL: Efficient Agentic Reinforcement Learning Systems for Large Language Models," *Workshop on Systems for Agentic AI (SAA)*, in conjunction with the ACM Symposium on Operating Systems Principles (SOSP), Oct. 2025.
- [2] Lukman Jibril Aliyu, Umar Sani Muhammad, Bilqisu Ismail, Nasiru Muhammad, Almustapha A Wakili, Seid Muhie Yimam, Shamsuddeen Hassan Muhammad, and **Mustapha Abdullahi**, "Evaluating Deep Learning Models for African Wildlife Image Classification: From DenseNet to Vision Transformers," *Deep Learning Indaba*, Jul. 2025.
- [3] Oluwole Fagbohun, Nwaamaka Pearl Iduwe, **Mustapha Abdullahi**, Adeseye Ifaturoti, and OM Nwanna, "Beyond traditional assessment: Exploring the impact of large language models on grading practices," *Journal of Artificial Intelligence, Machine Learning and Data Science*, vol. 2, no. 1, pp. 1–8, Feb. 2024.
- [4] **Mustapha Abdullahi**, "POSTER: An Artificial Intelligence of Things (AIOT) based Assistive Smart Glass for the Visually Impaired," Oct. 2021.

TALKS

- Kedro Databricks Deploy**, QuantumBlack AI Labs, London, U.K. **Dec. 2023**
Beyond Building Machine Learning Models , Data Science Nigeria, Nigeria **June 2021**

ACADEMIC SERVICE

Reviewer

- 2025 MLforSystems Workshop—NeurIPS
2025 Muslims in ML Workshop—NeurIPS
2025 Evaluating the Evolving LLM Lifecycle—NeurIPS

TEACHING EXPERIENCE

- Teaching Assistant**, CS-E4780 Scalable Systems and Data Management, Aalto University **Autumn 2025**
Research Advisor, CS-E4645: Research Project on Data Intensive Computing, Aalto University **Spring 2025**
Teaching Assistant, CS-E4780 Scalable Systems and Data Management, Aalto University **Autumn 2024**

PROJECTS

Evolving AlexNet CNN Maize Plant Disease Classification **July 2021 to Oct. 2021**

- Collaborated to propose a modified Binary Particle Swarm Optimisation (BPSO) algorithm to optimise the structure of the fully connected (FC) layers of CNNs.
- Developed a Tensorflow code implementation of AlexNet model
- Collaborated to derive an optimised model architecture achieving an improved accuracy of 98.3%.

iGlass **April 2020 - Oct. 2021**

- A smart assistive device for the visually impaired for reading out text, object detection, scene description, etc. using Raspberry Pi, AWS Rekognition, and Google Cloud AI.
- Spearheaded the development of iGlass prototype.

CompengBot **March 2021 to April 2021**

- A fast and efficient chatbot to answer queries of students and the general public related to the computer engineering department. Deployed on the departmental website and Telegram page.
- Easy and improved access to information about department and school.
- Quick access to details about courses and lecturers contact.

Transfer Learning for Classification of Cat, Dog and Pandas **Dec. 2020**

- Used densenet-161 as feature extractor and achieved an accuracy of 97.3%.

VOLUNTEERING

2025 Deep Learning Indaba **Feb. 2025 - Present** *Application Reviewer* *Remote*

Black in AI - Emerging Leaders in AI Grad Prep Program **Dec. 2024 - Present** *Academic Mentor* *Remote*

iScholars Initiative (iSI) **Sept. 2023 - Present** *Academic Mentor* *Remote*

2023 IEEE International Conference on Robotics and Automation (ICRA) **May 2023** *Volunteer* *London, U.K.*

Arewa Data Science Academy **Feb. 2023 - Present** *Deep Learning Mentor* *videos* *Nigeria, Remote*

Kada Hive Innovation and Tech Hub **May 2022** *Robotics Instructor - Play Tech Kids Programme* *Kaduna, Nigeria*

Data Science Nigeria **June 2021** *Data Science Tutor* *Zaria, Nigeria*

- Assisted participants in setting up data science software and packages.
- Supported the lead tutor in introducing participants to machine learning tools (such as numpy, pandas and scikit-learn).

Ahmadu Bello University **Feb. 2018 – April 2021** *Tutor - Control Engineering (COEN303, COEN407, COEN507)* *Zaria, Nigeria*

Computer Engineering, Ahmadu Bello University **Aug. 2019** *Instructor - Python Bootcamp for Kids* *Zaria, Nigeria*

TECHNICAL SKILLS

Programming Languages: Python, Scala, C++, MATLAB, $\text{\LaTeX}$, ROS

ML Frameworks: PyTorch, Tensorflow, Keras, Kedro, Apache-Spark, Scikit-Learn, XGBoost

Cloud Services: GCP, AWS, Databricks Platform, Huawei Cloud Platform

Operating System: Linux (Ubuntu, Cent OS 7.*), Windows

Big Data Tools: Hadoop, Apache Spark, Apache Hive, Hbase, MySQL, Huawei MapReduce Service, Gremlin

Microprocessors: Raspberry pi (zero, 3), Arduino, ESP32, ESP8266

Soft Skills: Leadership, Problem solving, Team Player, Time Management, Interpersonal Skills

PROFESSIONAL MEMBERSHIPS

- Black in AI, member
- Black in Robotics, member
- Nigerian Computer Society (NCS), member
- Data Science Nigeria, member
- Nigerian Society of Engineers (NSE), Graduate Member

HOBBIES

Watching anime; playing football; talks about AI and the future